GE Energy

GEK-SA-1051

Mark* VIe DCS for CCPP Brazi

Operation and Maintenance

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Issued: 100812

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Overview

System Design

The Mark VIe*Distributed Control System (DCS) provides reliable monitoring and control of the plant from a central location. Operators monitor and control the overall plant startup and shutdown using HMIs through pre-configured graphic screens.

The Mark VIe DCS consists mainly of the following redundant networks:

- Input/Output Network (IONet)
- Plant Data Highway (PDH)
- Unit Data Highway (UDH)

The controllers (processors) communicate with I/O modules using redundant IONet. These controllers also are connected through the UDH network to provide process data to operator stations and other equipment.

The PDH network provides connection to printers and other external devices as well as redundant connection to all operator stations, the engineering server, and the Historian. The DCS specification of this project includes:

- Five controllers
- 14 I/O cabinets
- Four control room servers
- Two Gateway Human-Machine Interfaces (HMIs)

And one of each of the following:

- Engineering workstation
- Historian workstation
- Large screen Human-Machine Interface (HMI)
- Supervisory station HMI
- Asset Management HMI
- NTP time server
- Configuration Management System (CMS) server
- Web view computer

In addition, the system communicates with external programmable logic controller (PLC) devices and the electrical protection relays network through a Modbus® link or Ethernet Global Data (EGD) communications. For more information, refer to the section *Interface with the Mark VIe Control System*.

Controller and I/O Cabinets

Controllers (processors) communicate with plant equipment, receive operator commands, process input data, and provide final output. Communication with plant equipment is achieved through I/O modules, installed in I/O cabinets at different locations throughout the plant. Dual controllers are provided for redundancy for boiler and other common controls.

Controllers and Distributed I/O Cabinets

CAse / LineUp	Cabinet Name	Description	Location
CA001	10CSA01GH001	HRSG1/HRSG2 controllers	Electrical building
CA002	10CSA02GH001	BOPE/BOPS controllers	Turbine building
CA003	10CSA03GH001	BOPM controllers	Turbine building
CA004	11CSA10GH001	I/O lineup (HRSG1)	GT1 area
LU001	11CSA10GH002	I/O lineup (HRSG1)	GT1 area
CA007	12CSA10GH001	I/O lineup (HRSG2)	GT2 area
LU002	12CSA10GH002	I/O lineup (HRSG2)	GT2 area
LU003	10CSA30GH002	I/O lineup (BOPE)	Electrical building
CA012	10CSA30GH001	I/O lineup (BOPE)	Cooling towers building
LU004	10CSA40GH001	I/O lineup (BOPM)	Cooling towers building
LU005	10CSA40GH002	I/O lineup (BOPM)	Gas compressor room
LU006	10CSA40GH003	I/O lineup (BOPM)	GT1 area
LU007	10CSA40GH005	I/O lineup (BOPM)	ST area
CA021	10CSA40GH004	I/O lineup (BOPM)	GT2 area
LU008	10CSA20GH001	I/O lineup (BOPS)	ST area
CA024	10CSA20GH002	I/O lineup (BOPS)	Electrical building
LU009	10CSA40GH007	I/O lineup (HRSG1/HRSG2/BOPM)	Electrical building

Workstations

HMI	Description	Location	Software Installed
CRM1_SVR CRM2_SVR CRM3_SVR CRM4_SVR	Operator Station 1 Operator Station 2 Operator Station 3 Operator Station 4	Main Control Room	Microsoft® Windows® XP Professional Microsoft Office 2007 CIMPLICITY* version 7.5 ControlST* software version 3.06
EWS1	Engineering Station	Office Room	Microsoft Windows XP Professional Microsoft Office 2007 CIMPLICITY version 7.5 ControlST software version 3.06
HST1	Historian	Main Control Room	PI Historian Tag Limit: PI-HST-TAG 10,000 Tags License OSI Client & Interface
LS_SVR	Large Screen Display Workstation w/ 65 in Display	Main Control Room	Microsoft Windows XP Professional Microsoft Office 2007 CIMPLICITY version 7.5 ControlST software version 3.06
GAT1 GAT2	Gateway Server 1 Gateway Server 2	Office Room Office Room	Microsoft Windows XP Professional Microsoft Office 2007 Triangle MicroWorks software ControlST software version 3.06
AMS1	HART® (AMS) Server	Office Room	Microsoft Windows XP Professional Microsoft Office 2007 Emerson® AMS software ControlST software version 3.06
SUP_SVR	Supervisor Station	Main Control Room	Microsoft Windows XP Professional Microsoft Office 2007 CIMPLICITY version 7.5 ControlST software version 3.06
CMS_SVR	CMS Server	Office Room	Microsoft Windows XP Professional Microsoft Office 2007 CIMPLICITY version 7.5 ControlST software version 3.06
WEB	Web View Computer	Office Room	Microsoft Windows XP Professional Microsoft Office 2007 CIMPLICITY version 7.5 ControlST software version 3.06
NTP1	Timer Server	Engineering Room	Meinberg® LANTIME GPS One IRIG-B Output

Operator Workstations provide the interface between operators and plant equipment. These workstations are equipped with dual LCD monitors and installed with GE proprietary licensed CIMPLICITY* software to display process parameters. The ControlST* software suite is also installed on these workstations, but operators typically are not expected to use those applications.

Engineering Workstation is mainly used for making configuration changes in the control system. It is equipped with dual LCD monitors. ControlST software is installed on this workstation for access to required configuration. The Engineering Workstation is configured with CIMPLICITY software, and can be used as an Operator Workstation to monitor and/or control plant processes based on user log-in privileges.

Historian Workstation is used for recording and archiving data available over the network. This workstation contains OSIsoft® PI software and OPC® interface to communicate with the Mark VIe control system.

Process measurements and other crucial information selected for archiving is configured in the Mark VIe control system through the ControlST software. This data becomes available for PI software through OPC interface using the WorkstationST* OPC and Historian services running on the Historian workstation.

Large-screen Display Workstation One workstation provides a large-screen (Sharp® 65 in) display. This workstation includes the same functionality as the operator workstations.

Gateway Workstations provide information available on the plant network from third-party devices using Triangle MicroWorks (TMW) software. The workstations feature one-way (simplex) communication. A GE proprietary WorkstationST service is installed on these stations, with OPC DA server enabled. These workstations are not configured with CIMPLICITY software. They provide no monitoring or control over plant equipment for the operator.

Highway Addressable Remote Transducer (HART®) Management Workstation is provided to communicate with HART enabled field devices. A third-party Emerson® AMS software suite is installed on the workstation to provide user interface with HART devices. To establish HART communication to the end device, a HART message service is installed. This service enables HART messages to be sent and received from AMS to the device.

Supervisor Station has the same configuration as the operator workstations. It is equipped with dual LCD monitors and installed with GE proprietary licensed CIMPLICITY software to display process parameters. The purpose of the Supervisor Station is to monitor and/or control plant process by site manager.

Configuration Management System (CMS) server is the designated HMI to store the master version of the project file. The CMS operates as a client for all HMIs that run, configure, edit, or view ToolboxST files. These HMIs are required to check-out files from the Master File Server for editing. ToolboxST* application is not installed on the CMS Server.

WEB View Computer is used to monitor plant process remotely through the Internet. The Web View Computer has the same configuration as an HMI, but does not have the privileges required to modify plant process. Plant process is in view only mode. It is equipped with a single LCD Display.

Time Server provides reliable time synchronization among all network devices when a Network Time Protocol (NTP) server is provided. When locked in with satellites, this device receives Global Positioning System (GPS) signals and broadcasts them to other devices over the network. All compliant network devices are configured to receive signals from the time server and to synchronize their internal clocks.

Analysis of Failure in Service

Devices used in Mark VIe DCS are continuously monitored for failure. In case of a malfunction or failure, an alarm is annunciated to the Alarm Viewer. Authorized individuals should follow guidelines provided in GE standard documents or in the maintenance section of this document to aid in problem resolution.

Interface with the Mark VIe Control System

The Mark VIe control system supports several industrial standards and can be configured to interface with a wide variety of control products. The Mark VIe DCS furnished for the Brazi site is configured with Modbus (TCP), Modbus (RTU), OPC, and Ethernet Global Data (EGD) interfaces.

Interface Type	HW Connection	System	Field Device
Modbus RTU	RS-485 Serial PSCA module	Digital Protection Relays	GE Multilin protection relays
Modbus TCP (Redundant)	Ethernet RJ-45 to PSCA module	Gas Filtering and Metering Station, Gas Compressor, Auxiliary Boiler, Demineralized Water System	PLC
Modbus TCP (Non-redundant)	Ethernet RJ-45 to PSCA module	Cond. Polish System, Waste Water Treatment, Emergency Diesel, Cooling Water Treatment	PLC
EGD	Ethernet RJ-45 from Unit Data Highway (UDH)	Compressed Air, Auxiliary Boiler, Pumping Station, WTP	GE Fanuc PLC

Health and Safety

For a list of general safety, electrical standards, and guidelines used for the design of Mark VIe control components, refer to *GEH-6721 Volume 1, Mark VIe Control System Guide, Chapter 4, Codes, Standards, and Environment*.

Operation

DCS operator workstations typically are located in a central control room in an environmentally controlled area. These workstations provide reliable plant operation by using special user interfaces configured with specific requirements.

Startup

The startup process brings each piece of equipment to a typical operating state so the control system becomes available for plant operation. Prior to initial startup of Mark VIe control equipment, special care should be taken to avoid any malfunction. General guidelines for major components must be followed together with other recommendations from the original equipment manufacturer (OEM).

Workstations

Each workstation is factory-installed with Microsoft Windows and GE proprietary software. The workstations are pre-configured with separate accounts for Operators, Maintenance personnel, and Administrators. An operator dialog box automatically displays after a successful operator logon. Graphics display shortcuts are provided on the main desktop. Before initial startup of any workstation, verify the following:

- The CPU, LCD monitors, keyboard, and mouse are placed properly at appropriate location.
- Both LCD monitors are connected to the CPU video port using special split cable (provided with each workstation).
- Keyboard and mouse are connected to CPU appropriately.
- Network cables are connected to the NIC ports available on the CPU.
- Power cables for CPU and LCD monitors are connected properly.
- All power sources are of the correct voltage per equipment specifications.

Cabinets

Mark VIe cabinets are located in an environmentally controlled area that meets equipment specifications. Each cabinet has specific requirements that must be met before initial powerup. The startup of controller and I/O cabinets should be performed by trained maintenance personnel. Refer to the section *Maintenance* for details on powering on a Mark VIe control cabinet. Before startup, verify the following:

- Cabinet must be grounded properly as per GE guidelines.
- Power cables must be connected properly.
- Network cables must be properly installed and connected to either the controllers (controller cabinet) or the network switches (I/O cabinet).
- All power sources are of the correct voltage per equipment specifications.

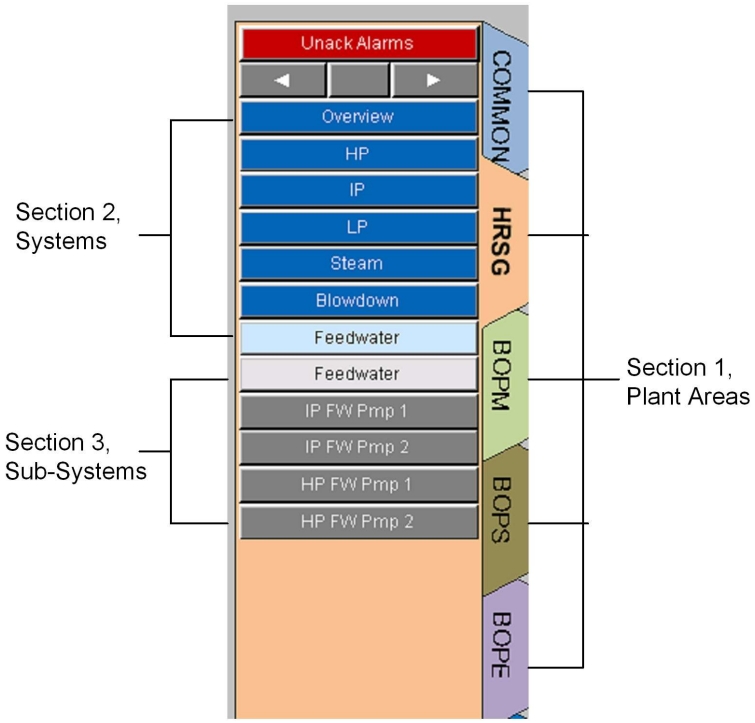
Operator Display Screens

The CIMPLICITY top banner is included on all graphic displays.



CIMPLICITY Top Banner

The Navigation Bar is available on all graphics displays, and it is used for quick navigation to operator screens. It contains three section: plant areas, systems, and sub-systems.

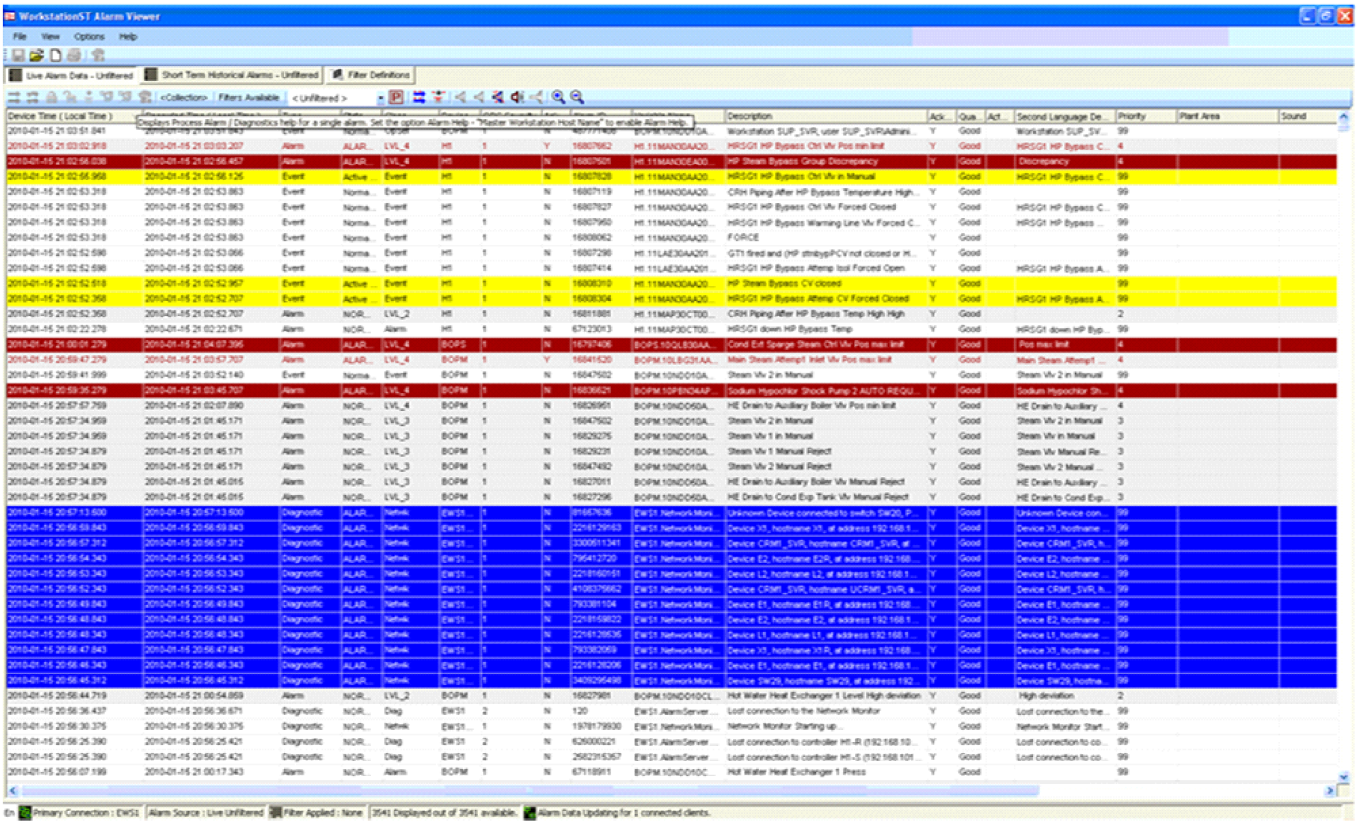


Navigation Bar

1. The first section defines plant areas such as *Common, HRSG, BOPM,* or *BOPS*.
2. The second section provides a selection of systems, for example *HP, LP, IP,* or *Feedwater*.
3. The third section is programmed for any displays showing details available within a selected system, such as *IP FM Pmp 1* and *IP FM Pmp 2*.

Alarm Display

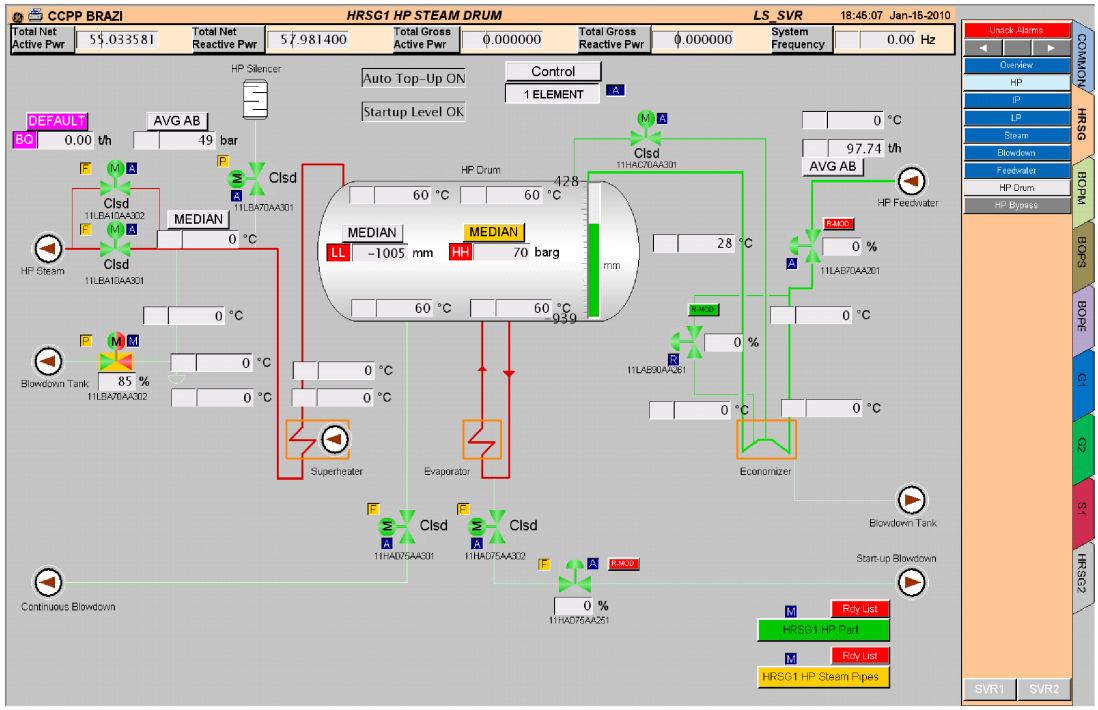
The alarm display is configurable using the Setup option. Filters can be created on devices (Unit), Alarm Class, or Alarm priorities. These filters also can be saved and re-loaded as needed.



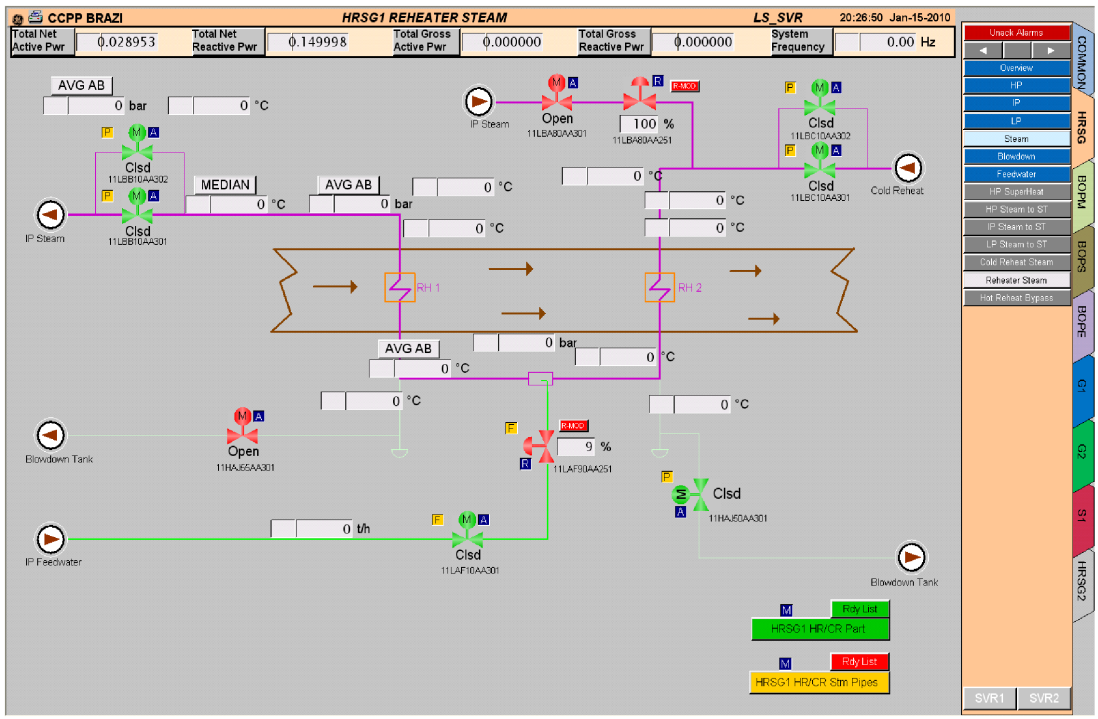
Alarm Display

Process Graphic Displays

Process graphics display the plant's process configuration and measurements, and provide operator control over these processes. These graphics use standard graphic symbols for devices and equipment connected together through lines similar to Process and Instrument diagrams (P&ID). Interactive objects further provide details for each piece of control equipment as pop-up faceplate graphics.



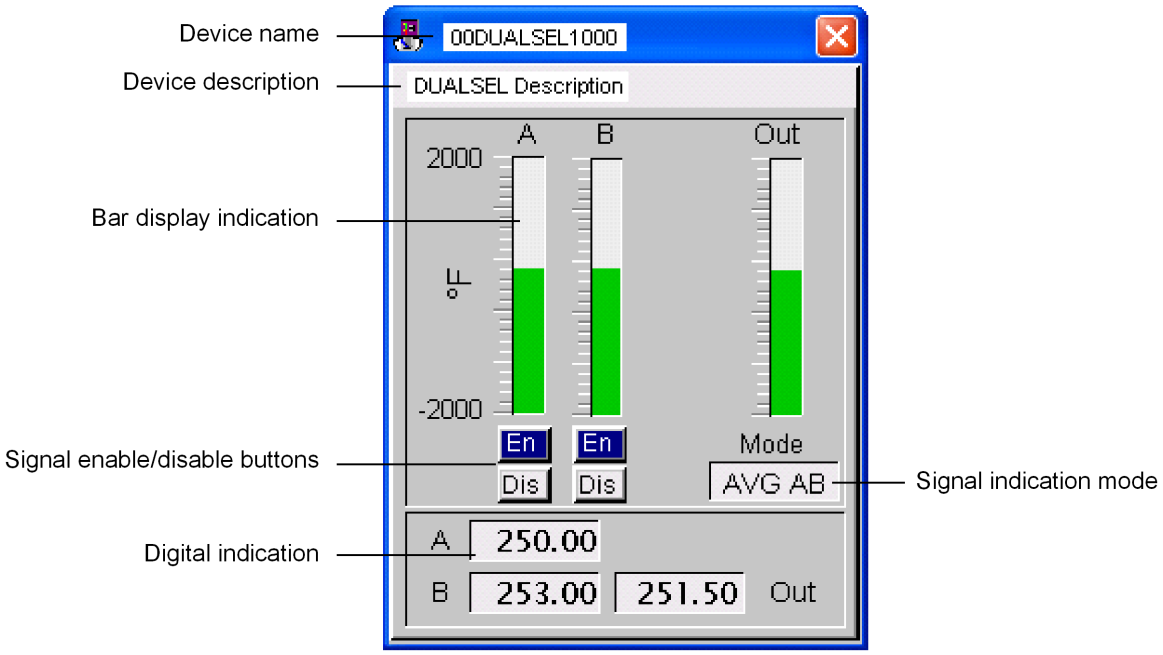
CIMPLICITY Operator Screen



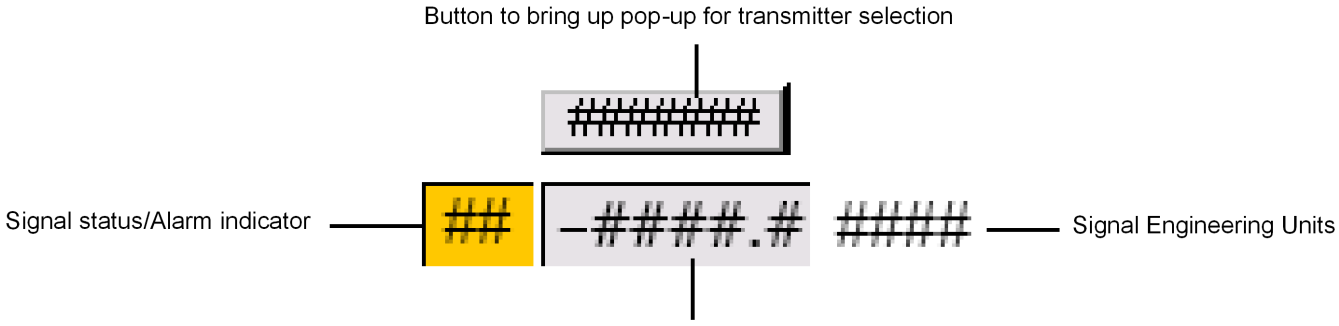
CIMPLICITY Operator Screen

Faceplate (Pop-up) Graphics

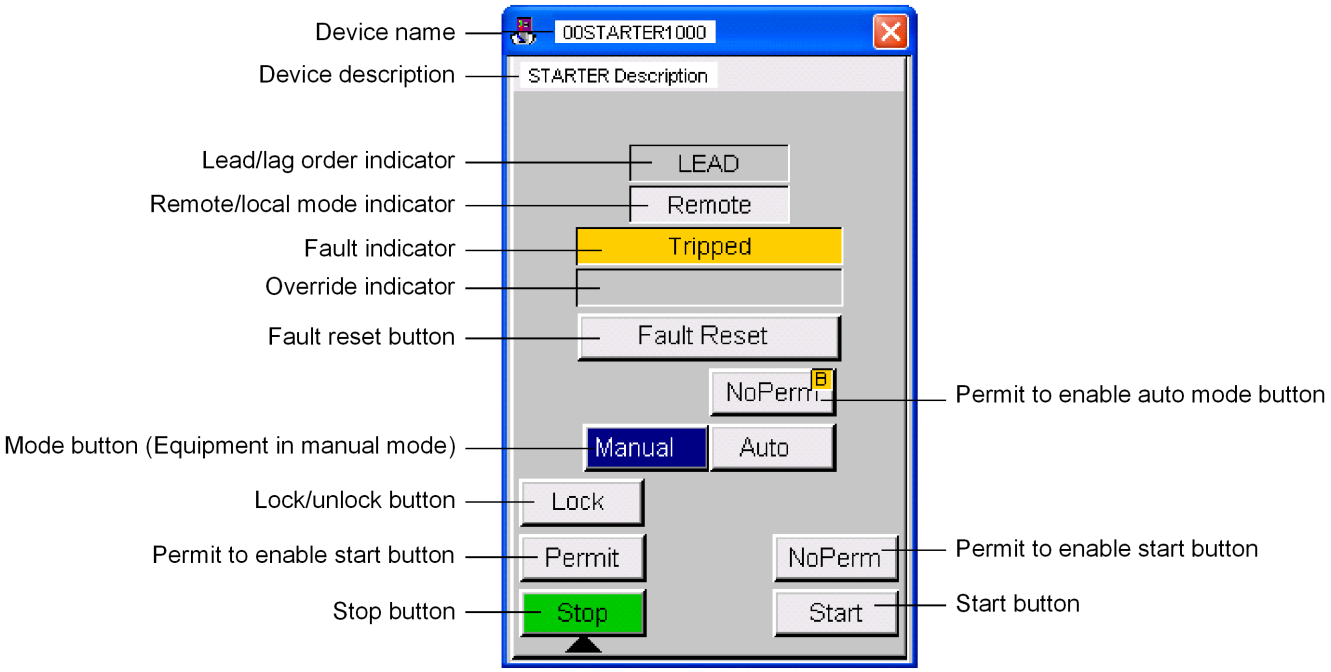
Operators can click buttons on objects to make changes. Some buttons display a pop-up of available choices.



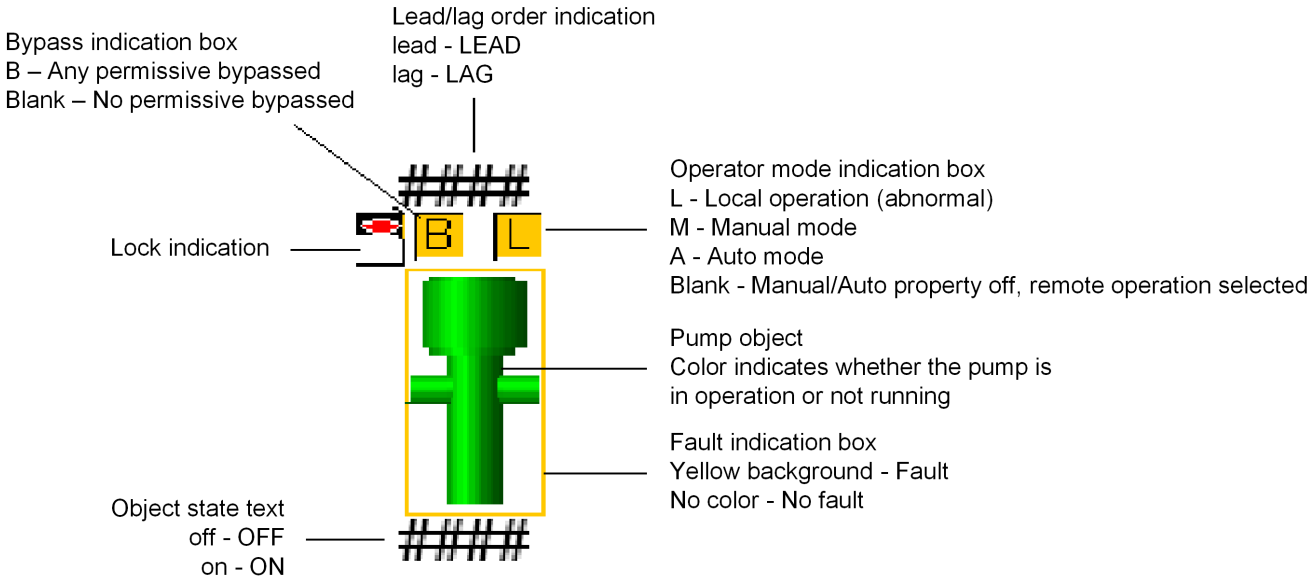
DualSel Faceplate



CIMPLICITY DualSel Object



Motor Faceplate



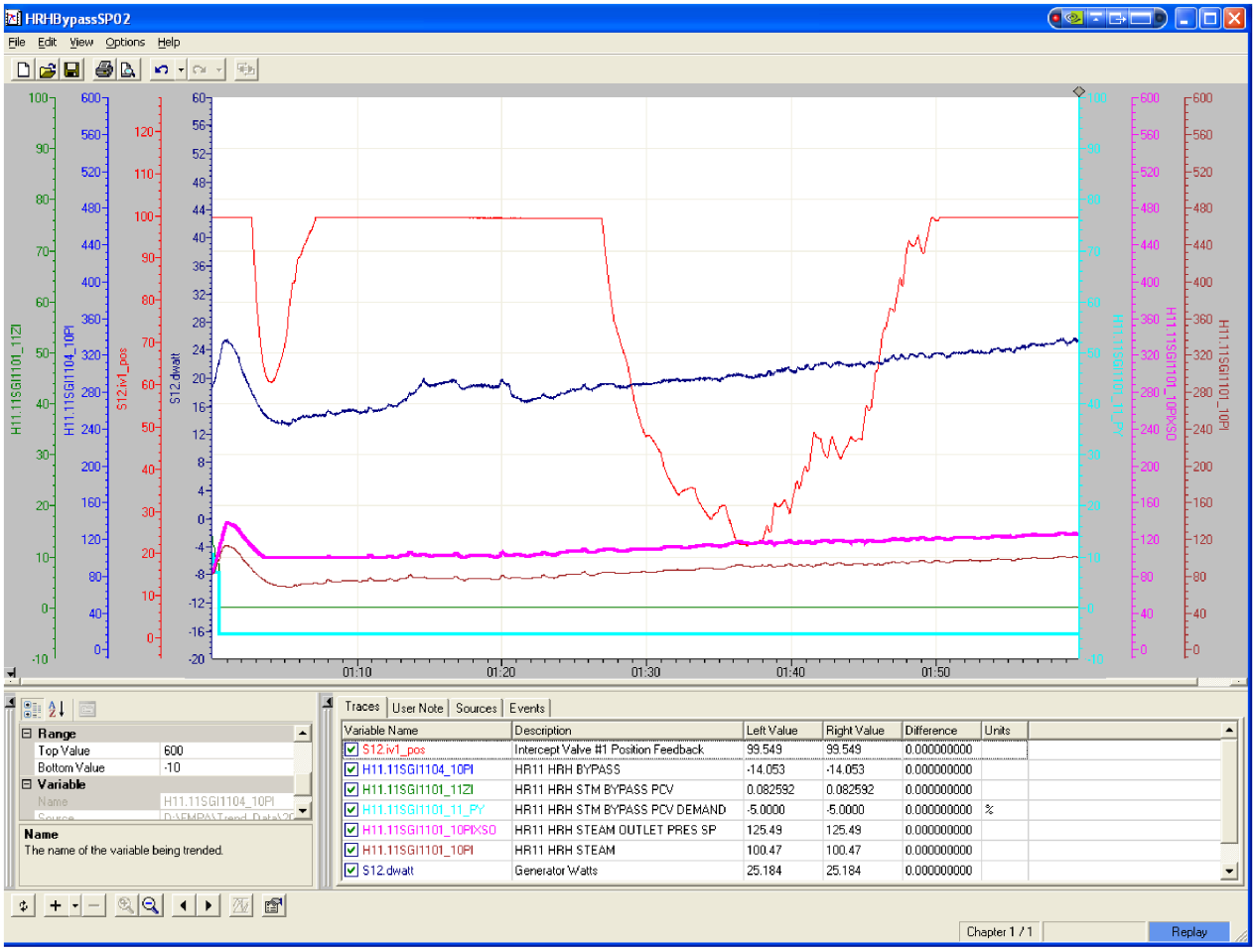
CIMPLICITY Pump Object

Trender

The ControlST software suite provides a powerful tool that captures and displays trend graphs of measurements and variable values in a real time system.

The Trender tool is accessed either from within the ToolboxST application or directly from the Windows Start menu. From the ToolboxST application, Trender is an option in the View menu of a component. If it is opened from the Start menu, files are stored with a **.tnd* or **.trend* extension.

The Trender window features several properties to format graphical data presentation. It includes different colors for each variable trace, adjustable scaling to fit several traces on the same axis, and user note that allow the user to save comments. Data captured through the Trender can be printed or exported to a **.csv* file.



Typical Trender Screen

Historian (Reports)

The Historian provided with the Mark VIe DCS is configured with Historian Report Package, a GE proprietary program. This program uses Microsoft Internet Explorer as the user interface. A desktop icon allows you to access these reports.

A set of varied reports are factory-programmed as guidelines for customer requirements. Shift reports, Daily reports, Monthly reports, and Custom reports can be configured with sets of variables available to the Historian. Timings for Shift reports can be changed to operator shift times, and can be setup to automatically generate and print.

Note Refer to *GEI-100752, Historian Report Configuration Instruction Guide* found on the Historian workstation computer as part of the *Historian Reports Package* software installation files (*Turbine_Historian_Report_Configuration.pdf*).

Shutdown

An operator workstation shutdown (for Maintenance and Administrator users only) is provided through Windows. The operator does not have workstation shutdown privileges, but can log off from the operator account.

Maintenance

Workstations

Refer to the Original Equipment Manufacturer (OEM) Documents section.

For further technical details, refer to section Workstations.

Each workstation includes the following:

- HP® xw4600 Workstation
- HP xw4600 80+ Energy Efficient Chassis
- HP xw4600 Workstation Localization Kit
- Microsoft Vista32B Downgrade to XP32B OS
- Intel® Core 2 Duo E6850 3.00 GHz, 4 MB, 1333 MHz FSB (Long Life)
- NVIDIA® Quadro NVS 290 256MB PCIe
- HP 4GB (4x1GB) DDR2-800 ECC RAM
- HP 80GB SATA 3Gb/s NCQ 7200 1st HDD
- HP 80GB SATA 3Gb/s NCQ 7200 2nd HDD
- HP 16X DVD+-RW SuperMulti SATA 1st Drive
- HP RAID 1 (Mirrored Array) Configuration
- HP No Floppy Disk Option
- HP USB Standard Keyboard
- HP USB Optical 3-Button Mouse
- HP ProtectTools Software
- Onboard Gigabit Ethernet NIC
- Broadcom® NetXtreme® Gigabit PCIe NIC [plus onboard NIC]

WorkstationST Service

There are several services configured to automatically start on a workstation after a normal startup.

There are several components to the WorkstationST service. Depending on the desired function of each workstation, WorkstationST features can be selected through configuration tools. Features include:

- EGD Configuration server
- Alarm server
- OPC AE server
- WorkstationST Alarm Viewer
- OPC DA server
- Recorder
- Historian
- NTP (Time Synchronization System)
- Network Status Monitor
- HMI configuration

HART Message Service (HMS)

The HART Message service (HMS) runs on AMS workstation and provides a bridge between Emerson AMS software and HART-enabled devices in the Mark VIe control system. The HMS server provides the following features:

- HMS Configuration Utility
- HMS Status Monitor
- HMS Status Viewer

Controllers

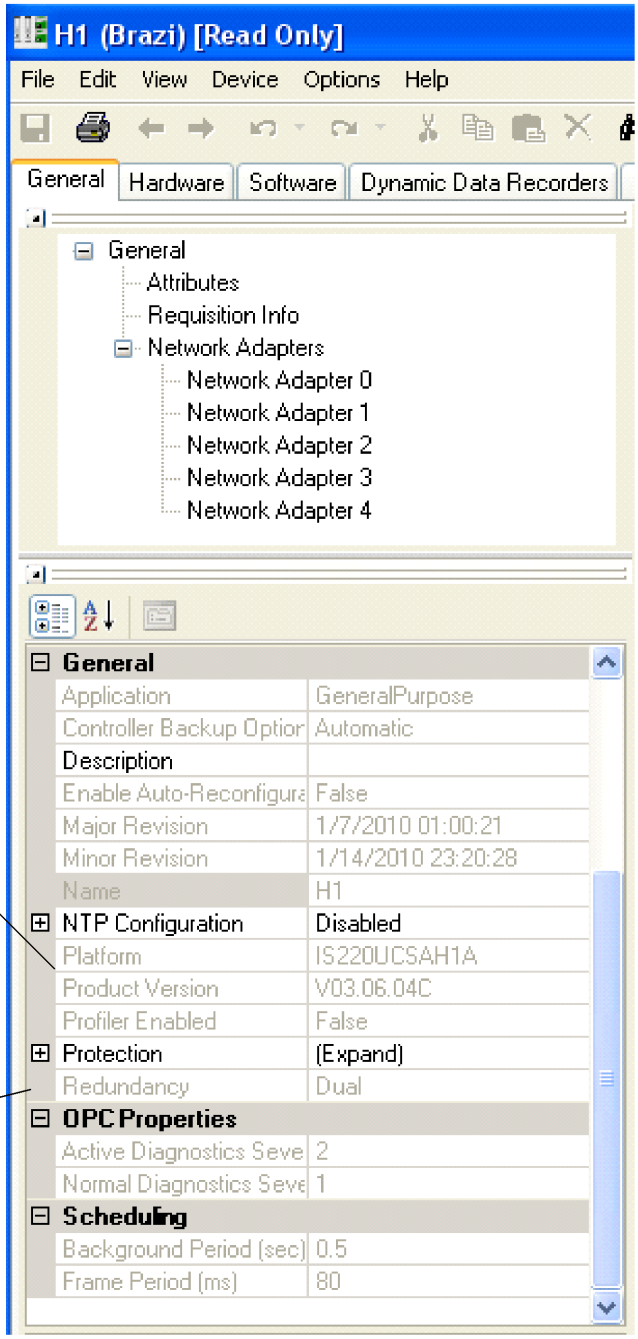
Refer to GEH-6721, Vol. 2, Mark VIe Control System Hardware Guide, section UCSx Controller.

The controllers used for BOP and HRSG controls are UCSA single board modules. The IS220UCSAH1+ Module contains a 667 MHz Power QUICC II Pro Freescale processor. Two 10/100BaseTX Ethernet ports provide connectivity to the UDH, and three additional 10/100BaseTX Ethernet ports provide connectivity to the IONets.

Note The new controller is required to undergo an initial setup and download procedure before it can be available for service.

➢ **To configure a new controller**

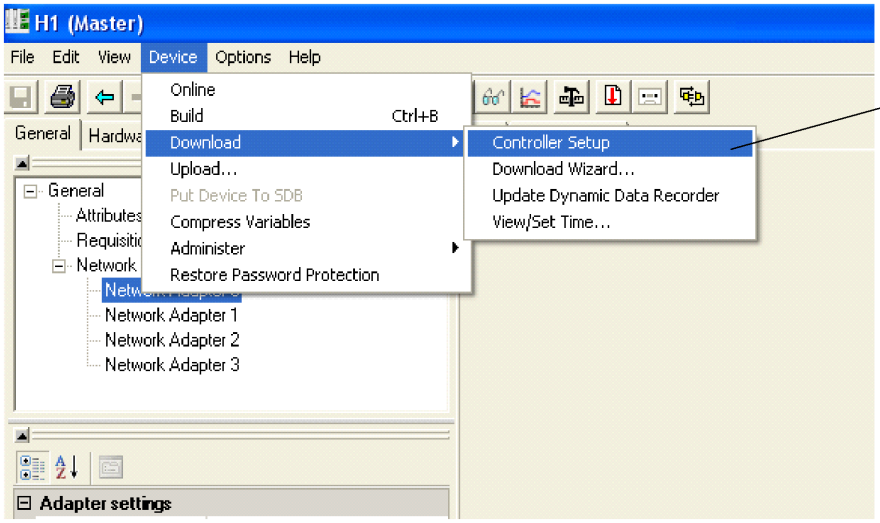
1. From the ToolboxST application open the **.tcw** file.
2. Double-click the controller. The **Component Editor** displays.



From the **General** tab **Property Editor**, verify that the **Platform** is set correctly for the selected controller.

Verify that the **Redundancy** is **Dual**.

3. Connect the Serial-to-Ethernet adapter and Ethernet cable to the serial port of your computer and connect to the COM port on the R Controller.



From the **Device** menu, select **Download** and **Controller Setup** to display the **Controller Setup Wizard**.

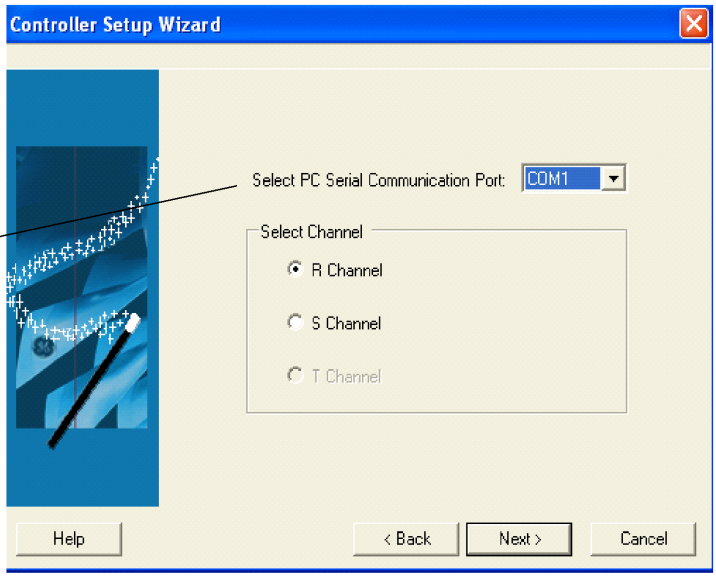
When the welcome wizard displays, click **Next**.

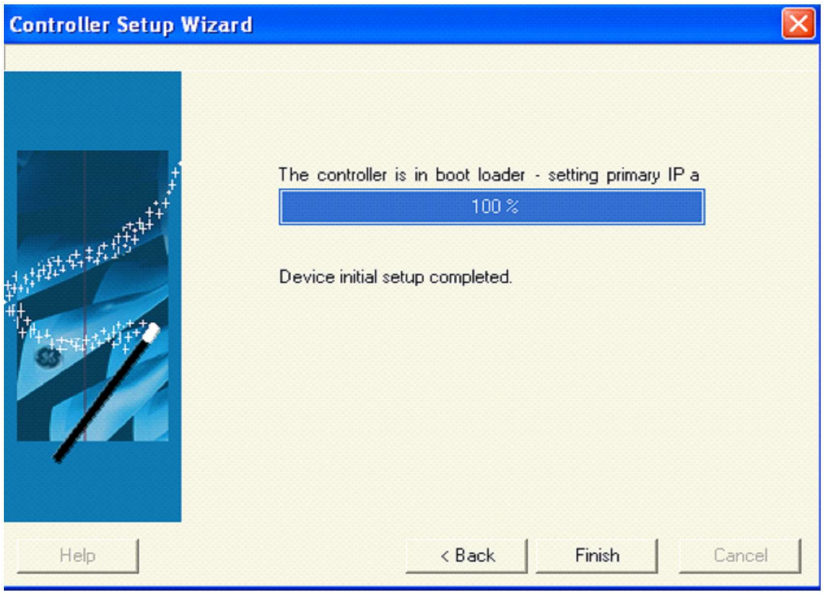
Select the **Configure Network Address** check box and click **Next**.

From the **Select PC Serial Communication Port** drop-down list, select the desired COM port, and verify that the COM port is set correctly.

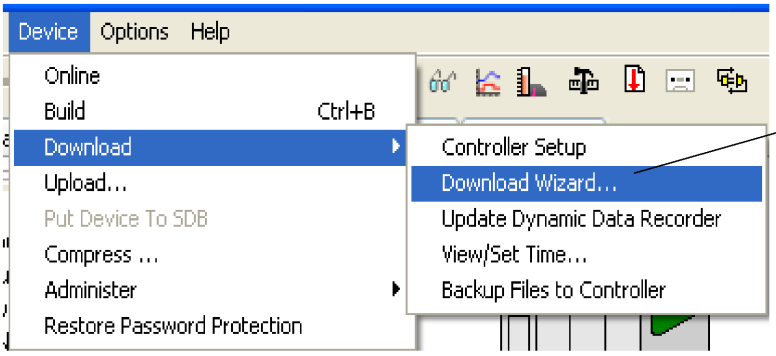
Click **Next**.

The initial setup begins.





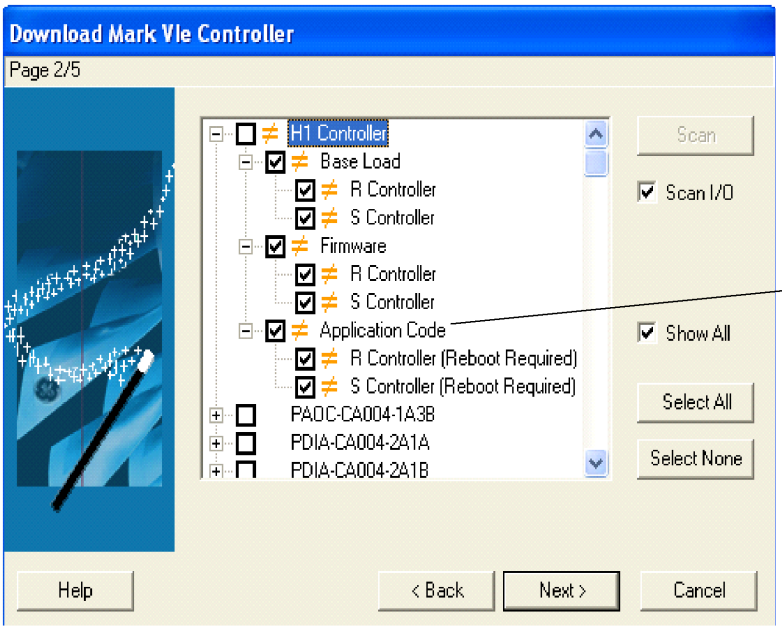
Click **Finish** and reboot the controller.



From the **Device** menu, select **Download** and **Download Wizard**.

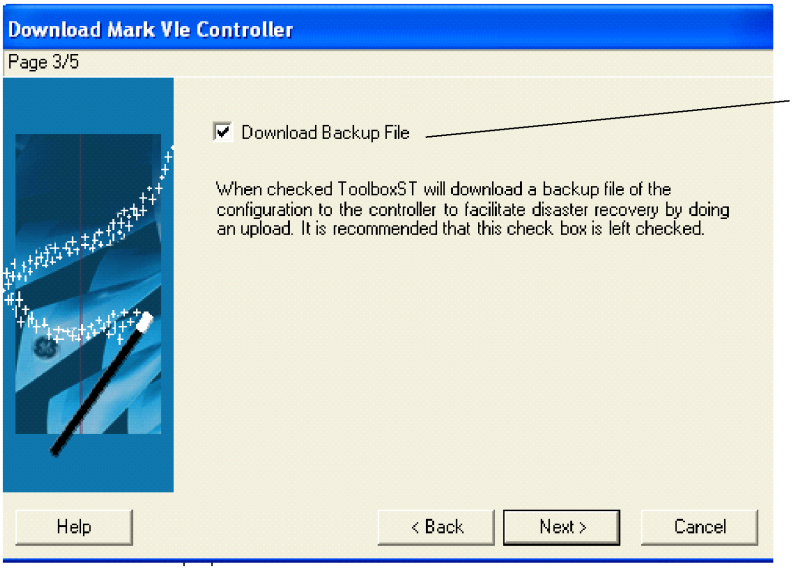
The **Download Mark VIe Controller** displays.

When the welcome wizard displays, click **Next**.



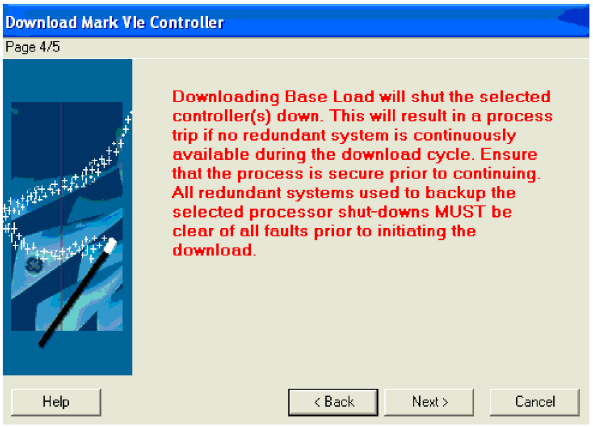
Select the **Base Load**, **Firmware**, and **Application Code** check boxes to download.

Click **Next**.



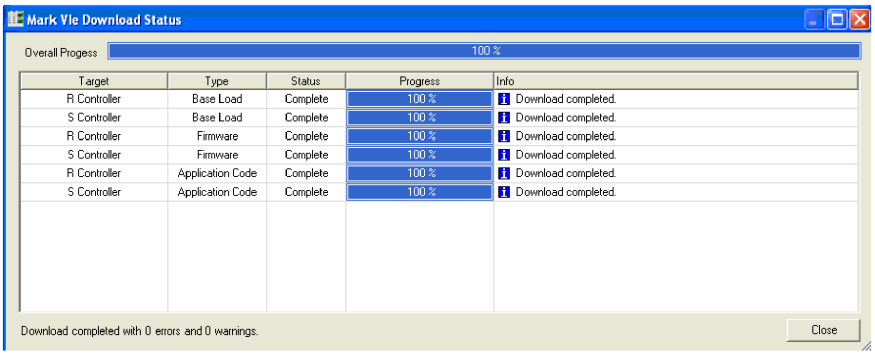
Select the **Download Backup File** check box.

Click **Next**.



Click **Next**.

4. If download prompts display, click **Yes**.



When the download is complete, the controller automatically restarts, and is available on the network for service.

I/O Cabinets

Refer to GEH-6721 Vol. I, Mark VIe Control System Guide, Chapter 7 Maintenance and Diagnostics.

I/O cabinets contain I/O modules, network switches, and power supplies. The I/O modules used in BOP and HRSG I/O cabinets are simplex modules. The major components in an I/O cabinet are continuously examined for their health with self-diagnostic routines. These diagnostics are available through the ToolboxST application for configuration, or for viewing current states. Diagnostic alarms are available through alarm windows on the operator display.

Network Switches

Network switches provided with the Mark VIe control are factory-programmed to provide reliable connectivity to the network devices. There are two types of network switches.

AT-8516F-VLAN	Allied Telesyn® AT-8516F/SC 16 port Fiber 100FX switch (SC connection) with two expansion bays. Rackmount hardware included. Configuration VLAN. For use in multi VLAN systems. Includes PI manufacturing IEC 6 ft power cord or equivalent.
AT-8624-VLAN1-PT	Allied Telesyn AT-8624T/2M-10 switch (port triggering), with 2 plug-in AT-A45/SC modules. Configuration 8624_Vlan_Tr.cfg + toggle.scp. For use in multi VLAN systems with port triggering. Includes PI Manufacturing IEC 6 ft power cord or equivalent

Note For component documents, refer to the section, Original Equipment Manufacturer (OEM) Documents.

Time Server

For component documents, refer to the section, Original Equipment Manufacturer (OEM) Documents.

A GPS receiver/time server is provided for time synchronization of network devices with a global satellite clock. Time server device specifications are as follows.

- GPS or IRIG-B Synchronized NTP Time Server (LANTIME/M300/MRS-TCXO-T/TC-1-1/ANT/50)
- NTP time server with free running clock, synchronized by IRIG-B or GPS with Antenna. Includes:
  - Down converted GPS antenna with 50 meters RG-58,
  - Antenna Cable capable of up to 300 meters cable run of RG-58,
  - 2 x LAN interface, RJ-45 connector, status LEDs for link, activity, speed (10/100Mbit),
  - 1 x RS-232 front panel interface, 9-pin D-Sub male connector for initial setup and configuration,
  - 1 x USD (Rev 1.1) front panel interface,
  - 1 x –LC display, 40 character x 2 rows,
  - 3 x bicolor LEDs; Ref. Time (e.g. GPS), Time Synchronization Service (NTP) and Network-Link status,
  - 1 x red alarm LED (configurable).
- Reference Inputs:
  - 1 x Meinberg GPS antenna input, BNC connector, isolated,
  - 1 x Time Code modulated input, BNC connector, isolated by transformer,
  - 1 x Time Code unmodulated input BNC connector, isolated by opto-coupler.
- Signal Outputs:
  - 2 x RS-232 interface, independent, 9-pin D-Sub female connector, with following formats: Meinberg Standard-Telegram, SAT, NMEA0183 (RMC), Uni Erlangen (NTP), COMPUTIME, SYSPLEX-1, SPA, RACAL,
  - 1 x Pulse Per Second (PPS), TTL into 50 Ω, pulse duration 200 msec, active high, BNC connector,
  - 1 x standard frequency 10 MHz, TTL into 50 Ω, female BNC connector,
  - 1 x IRIG-B amplitude modulated sinewave output, peak-to-peak voltage: 3 V into 50 Ω, BNC connector,
  - 1 x IRIG-B pulse-width modulated dc output (DCLS), TTL into 50 Ω, active high, BNC connector,
  - 1 x alarm relay output, change-over contact, 3-pin DFK connector.
- An NTP Time Server with:
  - 50 ft Cable with 50 Ω Connector,
  - 50 Ω T Connector,
  - (2) 50ft CAT6 Ethernet Cable Grey RoHS

Note For operating environmental conditions, general health and safety, electrical standards, and storage instructions, refer to *GEH-6721, Mark VIe Control, Volume I System Guide, Chapter 4 Codes, Standards, and Environment*.

Reference Documents

Original Equipment Manufacturer (OEM) Documents

This tables lists third-party technical documentation for equipment used in the Mark VIe DCS.

Equipment	Description	Document
Workstation	Hewlett-Packard (HP) Model xw4600 LCD Displays (HP) – HP L2045W (20 in Widescreen)	HPServiceAndTechRef.pdf HP LP2045W User Manual.pdf
Network Switches	AT-8624t-2m Series Switch (Allied Telesyn) AT-8516 (Allied Telesyn)	AT-8600 Series Switch Installation and Safety Guide.pdf AT8516_Web Browser Int User Guide.pdf
Power Supplies	TIS 600 W Power Supply (TRACO®)	TRACO TIS Power Supplies.pdf
NTP Time Server	GPS NTP Time Server	M300 GPS NTP Time Server Manual.pdf
IONet Network Switches	508FX, 509FX, 516TX, 517FX (N-TRON®)	N-Tron 500 Series Industrial Ethernet Switches Manual.pdf
Copper-to-Fiber-optic Converter	Copper-to-Fiber-optic Converter (Transition Networks®)	Transition Networks Copper_to_Fiber_Optic Conversion Manual.pdf
Printers	HP LaserJet 5200 B&W – 11x17 – 220 V HP LaserJet 3525X Color – 8.5x11 HP OfficeJet PRO8000 Color Inkjet – 8.5x11	HP LaserJet 5200 Printer Manual.pdf HP LaserJet 3525X Printer Manual.pdf HP OfficeJet PRO8000 Printer Manual.pdf
Large Screen Display	Sharp PN-G655RE LCD	Sharp PN-G655RE User Guide.pdf

Controls & Power Electronics, CoE Documents

GEH-6721 Mark VIe Control, Volume I System Guide

GEH-6721 Mark VIe Control, Volume II System Hardware Guide

GEH-6721 Mark VIe Control, Volume III Diagnostics and Troubleshooting Guide

GEI-100679 DCS HMI Graphics Application Guide

Note Site-specific installation drawings and bill of materials (BOM) are shipped directly to the project site.



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